Jatindar Kumar Mamnani

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PROFESSIONAL PROFILE

Skilled and Innovative Computer Vision Researcher with proven expertise in designing and implementing ML models for autonomous systems. Skilled in Python, PyTorch, and TensorFlow, with a strong track record in deep learning and multitask learning projects. Seeking opportunities to further advance machine learning capabilities in real- world applications.

EDUCATION

- BS in Computer Science School of Electrical Engineering and Computer Science , 2.93
- Matric / O levels , (2019)
- Intermediate / A levels , (2021) Bachelor of Science in Computer Science National University of Sciences and Technology , (2025)

WORK & INTERNSHIP EXPERIENCE

CodSoft

25-Jul-2024 - 25-Sep-2024

- Vision Research and development company (24-Jun-2024 - 24-Sep- 2024) • Conduct research on the applications of computer vision and data science in Autonomous cars • Design and implement ML models and algorithms • Assist in collecting and preprocessing relevant datasets • Contribute to the development and refinement of ML models and algorithms • Document findings, metho Communicate with managers to set up campus computers used on campus • Assess and troubleshoot computer problems brought by students, faculty and staff • Maintain upkeep of computers, classroom equipment, and 200 printers across campus

PROJECTS

Prompt-No-Code Website Assistant

- Developed an AI
- powered no
- code chatbot assistant that extracts contextual insights from website URLs for automated responses.
- Engineered an LLM
- driven chatbot system using FastAPI, enabling real
- time, context
- aware user interactions.
- Optimized API
- driven chatbot responses for accuracy,

automatio

- SemEval
- Task
- 8
- Table Answering
- Built a tabular QA system using transformer

- based NLP for structured data retrieval.
- Implemented text
- to
- SQL conversion using pre
- trained language models to generate structured queries.
- Conducted model evaluation using accuracy, precision, recall, and execution latency benchmarks.

Addressed NLU

- VQA
- Based
- Medical
- Advisor
- Developed a medical visual question
- answering system using CNNs for image features and LSTMs/Transformers for text queries.
- Implemented multimodal fusion to combine image and text features.
- Trained and evaluated on the VQA
- Med dataset, using accuracy, BLEU, and METEOR

metrics. • Tackled challen

- A Video Based Multi
- Task Learning Model for Autonomous Driving
- Developed a cutting
- edge multitask learning model for real
- time autonomous driving.
- Engineered to perform semantic segmentation, lane detection, and object detection on edge devices with high performance.
- Achieved 82.4% mIoU on the VIL
- 100 validation set for lane segmentation.

Demonstrated str

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TECHNICAL SKILLS

- Python socket programming
- Cuda
- Python and bash scripting
- Web scraping
- Computer vision
- Data science
- Flask
- Contrcat management

SOFT SKILLS

- Multi tasking
- Group work